

LESSON 7

Inference with frequency data

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3.1. Definition

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Basic concepts

Frequency table

Individual	Righ handed	Left handed
i=1	0	1
i=	1	0
i=1	1	1
...	1	1
i =I	1	0
Total	87	13

Application of the
proportions contrast

Contingency table

Sex	Righ handed	Left handed	Total
Male	43	9	52
Female	44	4	48
Total	87	13	100

Application of the
**independence and
homogeneity tests**

Inference with frequency data

The aim is to **draw conclusions** about the **population** values **of one or more probabilities**. These values are estimated in samples based on the relative proportions with which the corresponding events occur.

Proportions contrast - Definition

It is used to examine the **independence between two categorical variables**, based on the observed frequencies of the various combinations recorded in a contingency table

The aim is to use the results of a sample to determine whether the hypothesis that the probabilities p_i of the various events have certain pre-specified values p_{i0} in the corresponding populations is valid:

$$p_1 = p_{10} , p_2 = p_{20} , \dots , p_I = p_{I0}$$

To compare proportions, a test based on the **chi-squared distribution** is presented.

Proportions contrast - Definition

Hypothesis testing:

$$\begin{cases} H_0: p_i = p_{i0}, \forall i \\ H_1: \exists i / p_i \neq p_{i0} \end{cases}$$

Given a sample of N individuals, we consider a set of I mutually exclusive events $A_1, \dots, A_I$, with associated probabilities $p_{10}, \dots, p_{I0}$, whose union is equal to the sample space, such that:

$$\sum_i^I p_{i0} = 1$$

Proportions contrast - Definition

If the probability of an event A_i is indeed p_{i0} , then in a total of N observations, event A_i should occur on average $N \cdot p_{i0}$ times.

If H_0 is accepted, we denominate $N \cdot p_{i0}$ as theoretical value. In order to carry out the test, it is customary to require that these values be at **least equal to 5**.

If all the sample frequencies x_i are **similar** to the theoretical frequencies $N \cdot p_{i0}$, the null hypothesis H_0 will be accepted; whereas if some of the x_i **differ significantly** from their respective theoretical values $N \cdot p_{i0}$, the null hypothesis must be rejected.

What exactly do we mean by similar or very different?

Test Chi-Squared (χ^2)

A statistic d is proposed to measure the discrepancy between the observed values in the sample x_i and the theoretical values under the null hypothesis $N \cdot p_{i0}$.

$$d = \sum_{i=1}^I \frac{(\text{observed values} - \text{theoretical values})^2}{\text{theoretical values}} = \sum_{i=1}^I \frac{(x_i - N \cdot p_{i0})^2}{N \cdot p_{i0}}$$

d will be equal to zero if the observed values coincide exactly with the theoretical values, and will be higher the greater the difference between the two.

When H_0 is true, the d -statistic approximately follows a Chi-Squared distribution with $I - 1$ degrees of freedom (χ^2_{I-1}).

Test Chi-Squared (χ^2)

Hypothesis testing:

$$\begin{cases} H_0: p_i = p_{i0}, \forall i \rightarrow \textit{Accepted if } \chi_{I-1}^2 > d \\ H_1: \exists i / p_i \neq p_{i0} \rightarrow \textit{Accepted if } \chi_{I-1}^2 \leq d \end{cases}$$

H_0 will be accepted if the probability that a χ_{I-1}^2 value is greater than d is greater than the Type I risk (α) one is willing to accept, and will be rejected otherwise.

Contingency tables - Definition

The data collected in a study can be presented as a contingency table, in which the generic element x_{ij} indicates the number of times the value in **row i** has appeared in combination with the value in **column j** .

		$j = 1$	$j = 2$	
Sex		Righ handed	Left handed	Total
$i = 1$	Male	43	9	52
$i = 2$	Female	44	4	48
Total		87	13	100

$x_{12} = 9$

Contingency tables - Definition

Depending on whether the assignment of the observed individuals to one row or another in the table is random or not, two different tests are used:

Independence test: if rows order is not relevant.

Homogeneity test: if rows order is relevant.

Independence test

In a population, there are two defined qualitative random variables: the first has I levels and the second has J levels. The contingency table shows the observed frequencies for each of the $I \times J$ possible combinations of the two variables in a sample of N individuals selected at random from that population.

Hypothesis testing:

$$\begin{cases} H_0: I \text{ and } J \text{ are independent} \\ H_1: I \text{ and } J \text{ are **not** independent} \end{cases}$$

Independence test

If we denote by p_i the probability of the i -th category of the first variable, and by p_j the probability of the j -th category of the second variable, then the probability p_{ij} that an individual belongs to cell (i,j) of the table will be:

$$p_{ij} = p_i \cdot p_j$$

In case the null hypothesis (H_0) is accepted.

Independence test

In a sample of N individuals, the theoretical value that, on average, should have been observed in each cell, if H_0 is true, is $N \cdot p_i \cdot p_j$

The statistic d for measuring the discrepancy between the observed values x_{ij} and the theoretical values under the hypothesis of independence will be:

$$d = \sum_{i=1}^I \sum_{j=1}^J \frac{(x_{ij} - N \cdot p_i \cdot p_j)^2}{N \cdot p_i \cdot p_j}$$

where the parameters p_i and p_j must be replaced by their estimators p_i^* and p_j^*

Independence test

$$p_i^* = \frac{\sum_{j=1}^J x_{ij} \text{ (total row } x_i)}{N}$$

$$p_j^* = \frac{\sum_{i=1}^I x_{ij} \text{ (total column } x_j)}{N}$$

So that the expression of the expected “theoretical” values for each cell under the assumption of independence will be:

$$N \cdot p_i^* \cdot p_j^* = N \cdot \frac{x_i}{N} \cdot \frac{x_j}{N} = \frac{x_i \cdot x_j}{N}$$

Hypothesis testing:

The statistic χ^2 follows a Chi-Squared distribution with $(I - 1)(J - 1)$ degrees of freedom.

$$\begin{cases} H_0: I \text{ and } J \text{ independent} \rightarrow \text{Accepted if } \chi^2_{I-1, J-1} > d \\ H_1: I \text{ and } J \text{ **not** independent} \rightarrow \text{Accepted if } \chi^2_{I-1, J-1} \leq d \end{cases}$$

Homogeneity test

A problem that appears similar but is in fact of a different nature arises when the rows do not represent different, randomly selected alternatives of a qualitative random variable, but rather different treatments applied to individuals in the population or **samples collected under different conditions.**

The aim is to test the hypothesis that the different treatments (or the different samples) are **homogeneous with respect to the characteristics represented by the columns.**

Homogeneity test

The statistic d to test the hypothesis of homogeneity Will be:

$$d = \sum_{i=1}^I \sum_{j=1}^J \frac{(x_{ij} - t_{ij})^2}{t_{ij}}$$

where t_{ij} is exactly the same as that obtained in the case of the independence test:

$$t_{ij} = \frac{x_i \cdot x_j}{N}$$

Homogeneity test

Hypothesis testing:

The statistic χ^2 follows a Chi-Squared distribution with $(I - 1)(J - 1)$ degrees of freedom.

$$\begin{cases} H_0: I \text{ rows are homogeneous} \rightarrow \text{Accepted if } \chi^2_{I-1, J-1} > d \\ H_1: I \text{ rows are **not** homogeneous} \rightarrow \text{Accepted if } \chi^2_{I-1, J-1} \leq d \end{cases}$$

Exercise 1

- a) When a die was rolled 50 times, the result was 1 ten times, 2 five times, 3 eight times, 4 eight times, 5 seven times, and 6 the remaining 12 times. Is it reasonable to assume that the die is symmetrical?

- b) In crosses between white and black mouse hybrids, 15 white mice and 25 black mice were produced. Are these results consistent with the hypothesis that colour inheritance in mice follows a simple Mendelian pattern, implying theoretical ratios of $1/4$ for the recessive trait (white) and $3/4$ for the dominant trait (black)?

Exercise 2

A health questionnaire was administered to 125 people. Among the questions, they were asked whether they consumed alcohol and tobacco on a daily basis. Of the 55 people who stated that they smoked daily, 5 of them also consumed alcohol. Among non-smokers, 2 reported consuming alcohol, whilst 68 did not. Analyse whether there is a relationship between tobacco and alcohol consumption.

Exercise 3

Three treatments for a particular illness were compared, yielding the results shown in the table below:

Sex	Recovered	Non-recovered
Treatment 1	25	12
Treatment 2	15	18
Treatment 3	11	10

Can it be said that there are differences in effectiveness between the three treatments?
If so, between which ones?

Exercise 4

A company suspects that there are certain times of day when workplace accidents are more frequent. To investigate this phenomenon, they recorded the workplace accidents suffered by employees by time of day over the course of a year. The results are shown in the table.

Hours	8-10 h	10-12 h	13-15h	15-17h
Nº of Accidents	47	52	57	63

- Determine the type of test or comparison to be carried out and the null and alternative hypotheses.
- For a Type I risk of 1%, this examines whether the company's safety managers should determine whether there are specific time slots when accidents are more likely to occur or whether, on the contrary, they occur entirely at random.

Exercise 5

A car manufacturing company receives screws from various suppliers to secure the seats to the chassis.

To meet the specified quality standards, the screws must have an average length of 100 mm and a maximum standard deviation of 5 mm, to which all suppliers have committed by contract. As part of the company's quality control procedures, 10 screws from one supplier were sampled, and a standard deviation of 6 mm was obtained.

- a) Is the supplier meeting the terms of the contract?
- b) What is the approximate value of the Type II risk if the supplier's actual standard deviation were 9 mm?

Exercise 5

The company manufactures three different car models and sells them in Europe, the Americas and Asia. In order to make decisions about future production, it is necessary to analyse the preferences of each market to determine whether they differ by continent. To this end, 100 cars destined for each market have been selected, and the sales frequency of each model has been analysed.

	Model A	Model B	Model C
Europe	60	19	21
America	62	23	15
Asia	51	34	15

- c) What type of statistical test would you use, and what null and alternative hypotheses would you formulate?
- d) Determine whether the model preference depends on the continent.

Exercise 6

In order to determine the influence of the mounting position of electronic equipment on the incidence rate of two different types of faults that may occur in such equipment, a study has been carried out to record the number of units according to the three positions in which they can be manufactured and the type of fault they exhibit.

	Fault A	Fault B
Position A	30	15
Position B	40	45
Position C	20	12

- What type of test would be used? Justify your answer.
- Determine, for a first-order risk of 10%, whether or not there is a relationship between the types of failure and the installation position.